Do More Indebted Governments Pay Higher Yields? Government Debt and 10-Year Bond Yields in 24 OECD Countries, 2000–2024

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Repository: https://github.com/haojin-teng/econ3916-midterm **Notebook:** https://colab.research.google.com/github/haojin-teng/econ3916-midterm/blob/main/econ3916_midterm.ipynb

Question	Across 24 OECD countries from 2000 to 2024, is higher general government gross debt (% of GDP) associated with a higher 10-year government bond yield (%), holding inflation, real GDP growth and the current account balance constant?
Hypothesis	More debt raises the risk of default or of later inflation, so investors should demand a higher yield: I expect a positive coefficient on debt.
Headline finding	In the same year and with the same controls, 10 percentage points of GDP more debt go with a yield 0.023 percentage points higher (95% CI −0.020 to +0.067; −0.133 to +0.180 with errors clustered by country; N = 599). Within a country over time, the association is +0.278 (clustered 95% CI +0.188 to +0.368). Both are conditional associations, not causal effects.
Data	OECD 10-year benchmark yields via FRED (monthly, averaged by year) and the IMF World Economic Outlook, October 2026; 2000–2024; one row is one country-year; N = 599.
Method	OLS of the yield on debt, inflation, growth and the current account with year dummies (main model), HC3 robust standard errors; a second model adds country dummies.

1. The Question and the Data

Across 24 OECD countries from 2000 to 2024, is higher government debt associated with higher 10-year government bond yields, holding inflation, the current account balance and real GDP growth constant? The 10-year yield sets the cost of new long-term debt, so the answer matters to finance ministries deciding how much to borrow. I expect a positive coefficient, because more debt raises default and inflation risk.

Data. Yields are the OECD's 10-year benchmark government bond yields, compiled from national sources for its Main Economic Indicators and retrieved from FRED (IRLTLT01…M156N); I average each year's monthly values. General government gross debt and the current account (% of GDP), real GDP growth and CPI inflation come from the IMF World Economic Outlook, October 2026, compiled by IMF staff from national statistics; debt definitions differ slightly across countries, and recent years can be staff estimates.

Sample. The sample contains the 24 OECD members whose FRED yield series starts by January 2000. Iceland is left out: its series averages 4.30% in 2008 and 2009, when inflation was 12.7% and 12.0%, so it does not measure a nominal yield. This differs from my checkpoint post, which proposed 22 countries; the rule above replaces that list.

Missing data. The 13 members without a series from 2000 likely have higher yields, since a 10-year benchmark needs a liquid long-term market (missing not at random, MNAR); the results describe established bond markets. US debt for 2000 is missing because of WEO coverage, unrelated to yields (missing completely at random, MCAR), so I drop that row. Greece's July 2015 yield is missing because its banks were closed (MNAR); I keep the 11-month average. Table 1 summarizes the 599 country-years.

Table 1. Summary statistics, 24 OECD countries, 2000–2024

Variable (units)	Mean	Median	SD	Min	Max	N
10-year government bond yield (%)	3.2	3.2	2.3	−0.5	22.5	599
General government gross debt (% of GDP)	73.8	65.8	42.0	7.4	228.8	599
Current account balance (% of GDP)	1.2	1.2	5.5	−20.7	27.8	599
Real GDP growth (%)	1.9	2.0	3.0	−10.9	24.6	599
CPI inflation (%)	2.2	2.0	2.1	−1.7	17.1	599

Note: Sources: OECD via FRED (yields, retrieved 2026-11-08); IMF World Economic Outlook, October 2026 (all other variables). Lowest and highest values: yield, Switzerland 2020 and Greece 2012; debt, Luxembourg 2002 and Japan 2020; current account, Ireland 2019 and Norway 2022; growth, Spain 2020 and Ireland 2015; inflation, Ireland 2009 and Hungary 2023. Ireland's 2015 growth and 2019 current account reflect multinational firms' accounting.

2. Descriptive Evidence

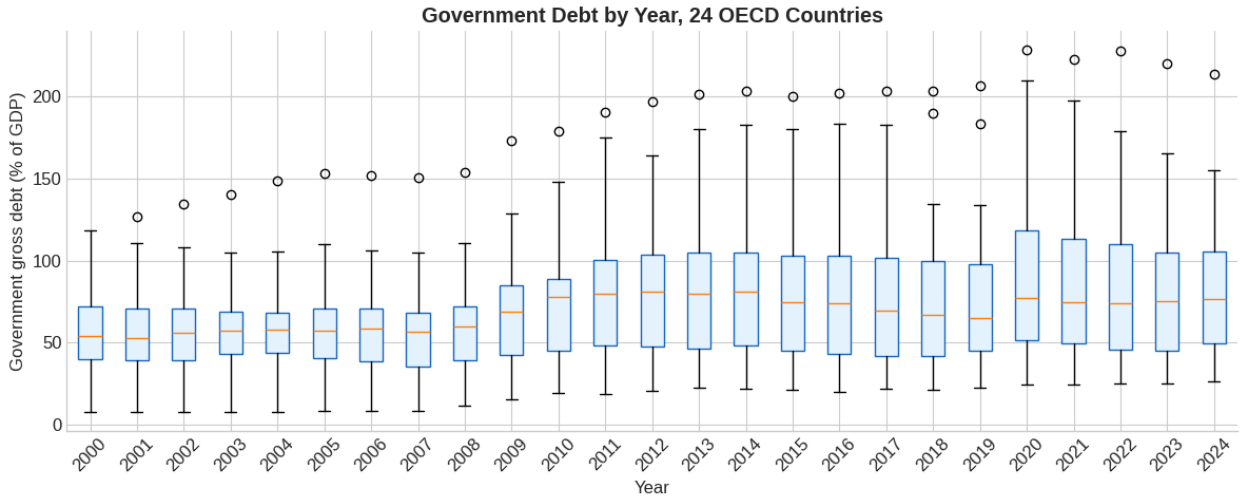


Figure 1. Government debt rose. The median country's debt rose from 54% of GDP in 2000 to 77% in 2024; boxes show the middle half of the 24 countries. Source: IMF World Economic Outlook, October 2026.

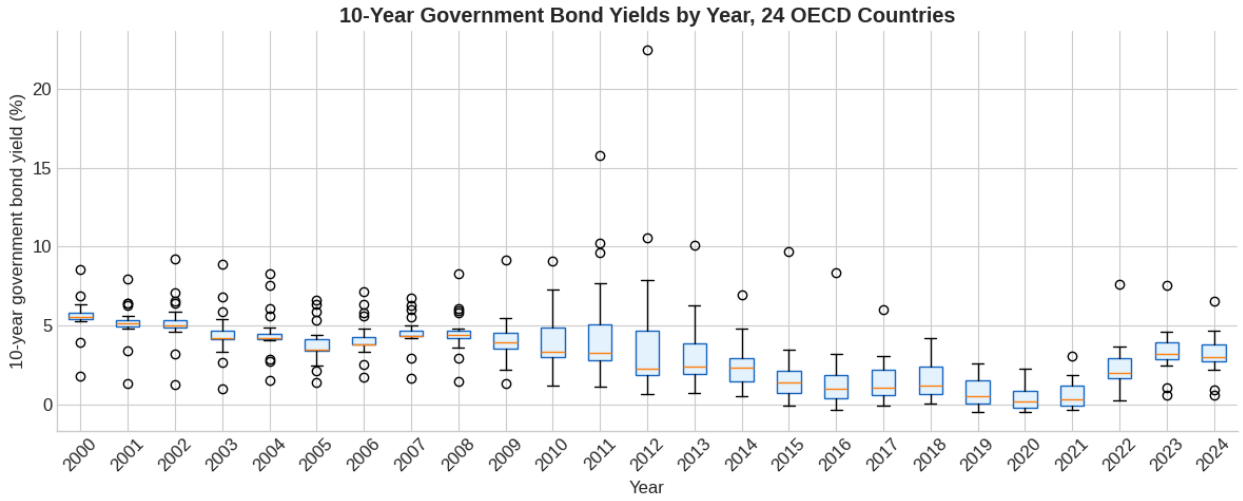


Figure 2. Yields fell until 2021, then rose. The median yield fell from 5.5% in 2000 to 0.3% in 2021 and was 3.0% in 2024. Source: OECD via FRED.



Figure 3. Within a year, two groups pull in opposite directions. Debt and yields relative to each year's cross-country mean, with one least-squares line through all country-years (+0.05 points of yield per 10 points of debt): Japan (top right) lies below it, the 2010–2012 crisis countries (bottom left) above it. Sources: as Figures 1 and 2.

Debt and yields trended in opposite directions, so the pooled correlation is −0.06 but +0.13 within years; the model therefore includes year dummies. The yield is right-skewed (skewness 1.57; Figure A1, Appendix), and Table 1 reports its median beside the mean. Because 30 country-years are at or below zero, a log is not possible, so the yield stays in levels; Section 4 checks how far the high-yield tail moves the estimate.

3. The Model

Table A1 (Appendix) reports OLS estimates with HC3 robust standard errors: column (1) without year dummies, (2) the main model, (3) with country dummies, and (4) without growth and the current account. Column (2), for country *i* in year *t*, is

$$Yield_{it} = \beta_0 + \beta_1\, Debt_{it} + \beta_2\, Inflation_{it} + \beta_3\, Growth_{it} + \beta_4\, CA_{it} + \gamma_t + u_{it}$$

where γ_t is a year dummy (base year 2000), so β_1 compares countries in the same year.

Debt. In the same year and with the same inflation, growth and current account, a country with debt 10 points of GDP higher has a yield 0.023 percentage points (2 basis points) higher (95% CI −0.020 to +0.067; p = 0.290). With errors clustered by country, the interval is −0.133 to +0.180. The association is small: one standard deviation of debt (42 points of GDP) goes with a yield only 0.10 points higher.

Inflation. Holding debt, growth, the current account and the year fixed, inflation 1 percentage point higher goes with a yield 0.305 percentage points higher (95% CI 0.199 to 0.412), well below one-for-one, since a 10-year yield prices a decade of expected inflation.

Across columns. Without year dummies the debt coefficient is −0.067; with country dummies it is +0.278 (standard error 0.038): within a country, years of high debt are years of high yields. Without growth and the current account it is +0.079.

Controls. Figure 4 shows the causal graph I assume, and the table gives each control's role. No control is a collider in this graph, so none opens a path.

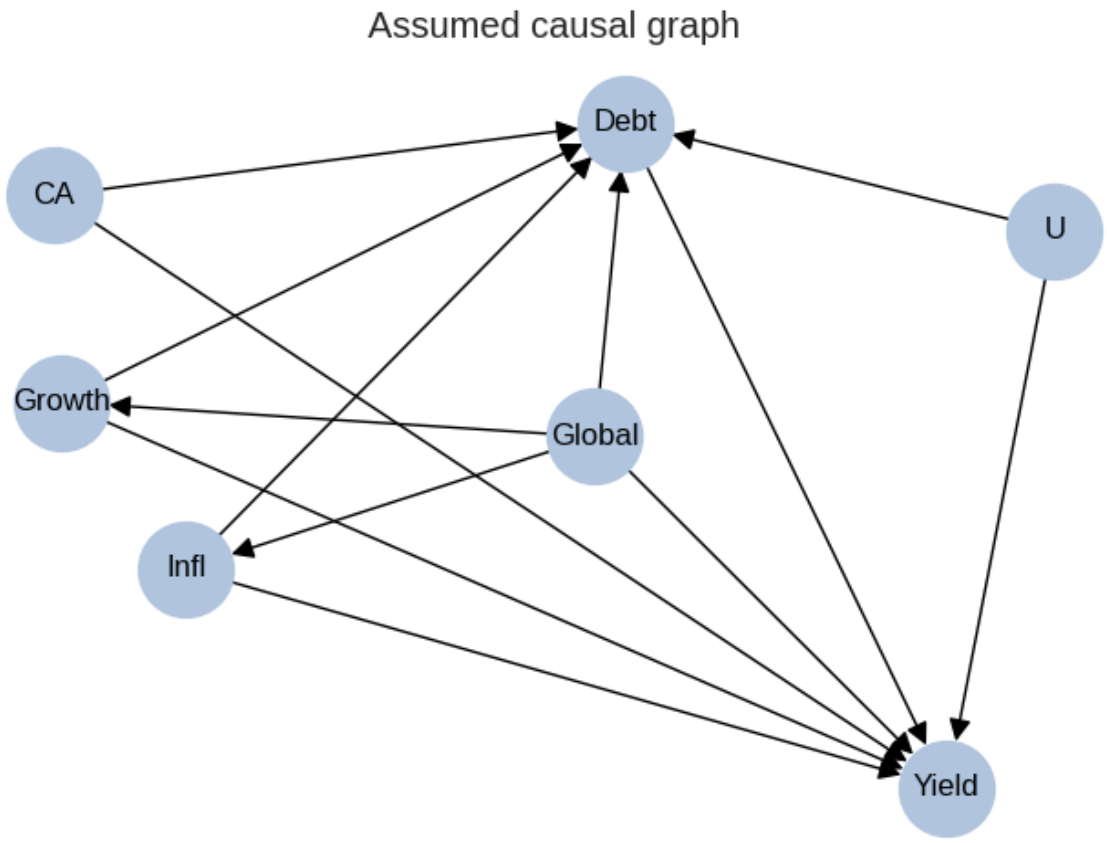


Figure 4. The causal graph I assume. Global: shocks common to all countries in a year (year dummies). U: fixed country traits such as the investor base (country dummies).

Control	Path it blocks	Bad control if
Year dummies	Common shocks to all countries (falling rates to 2021, 2008, 2020)	Not applicable
CPI inflation	Raises yields and lowers the debt ratio	Debt causes inflation
Real GDP growth	Lowers the debt ratio and raises real rates	Debt lowers growth (a mediator)
Current account balance	Saving surpluses lower yields and fund debt at home	Deficits cause current-account deficits
Country dummies, column (3)	Fixed traits: investor base, own currency	It turns the comparison into a within-country one

4. Diagnostics

Residuals. Residuals from column (2) average within 0.23 points of zero across fifths of fitted values and of debt, except for Japan (−2.11) and the 2010–2012 crisis countries (+3.26) (Figure A2).

Standard errors. The residual standard deviation rises from 0.93 to 2.15 points across fifths of fitted values (Breusch–Pagan LM = 81.3, $p < 0.001$), so I use HC3 standard errors. Clustering by country makes the debt standard error in column (2) 3.4 times larger (95% CI −0.133 to +0.180); in column (3) the interval stays above zero (+0.188 to +0.368).

Collinearity. VIFs are at most 2.73 in column (2), so collinearity barely matters there: debt's standard error is 1.17 times what it would be with uncorrelated regressors. In column (3) debt's VIF is 9.51 (standard error ×3.08), because country dummies absorb most differences in debt; the estimate remains precise.

Influential observations. No Cook's distance exceeds 1; the largest is Greece 2012 (0.348). Column (2) is fragile: +0.126 without Japan and −0.020 without Greece. Column (3) stays between +0.231 and +0.278 when Japan, Greece, the crisis years or the most influential country-years are dropped.

5. Threats and Conclusion

Is the main coefficient causal? For it to be causal, nothing else that moves yields could vary with debt once the year and the controls are fixed. That fails here, for four reasons.

- **Unmeasured differences.** Japan's domestic investor base and central bank keep its yields low; country dummies remove only fixed differences.
- **Central-bank bond buying.** Purchases, including the Bank of Japan's yield cap, lowered yields as debt rose.
- **Reverse causality.** High yields raise interest costs, and crises raise debt; without the crisis years column (3) is still +0.245.
- **Dependent observations.** Errors are correlated within countries and across euro members; clustering by country covers only the first.

Growth and the current account may also be mediators: dropping them (column 4) raises the estimate to +0.079.

Conclusion. Both estimates are conditional associations, not causal effects: column (2) compares countries in the same year, and column (3) compares years within a country. Across countries in the same year, 10 more points of debt go with yields only 2 basis points higher (95% CI −0.133 to +0.180 points with errors clustered by country). Within countries, they go with yields 0.278 points higher (clustered 95% CI +0.188 to +0.368), robust to dropping Japan, Greece or the crisis years. A reader can be fairly confident that the within-country association is positive, but not that debt matters across countries in the same year. With another month, I would allow for correlation across euro members, use start-of-year debt and add central-bank bond holdings.

References

Federal Reserve Bank of St. Louis. Interest Rates: Long-Term Government Bond Yields: 10-Year: Main (Including Benchmark), series IRLTLT01…M156N for 24 OECD countries, from OECD Main Economic Indicators. FRED. Retrieved 2026-11-08.

International Monetary Fund. World Economic Outlook, October 2026: general government gross debt (GGXWDG_NGDP), current account balance (BCA_NGDPD), real GDP growth (NGDP_RPCH) and CPI inflation (PCPIPCH), through the IMF DataMapper API. Retrieved 2026-11-08.

Appendix

Table A1. 10-year government bond yield regressed on government debt and controls

	(1) No year dummies	(2) Main	(3) Country dummies	(4) No growth or CA
Debt (per 10 pp of GDP)	−0.067	0.023	0.278	0.079
	(0.024)	(0.022)	(0.038)	(0.030)
CPI inflation (%)	0.242	0.305	0.052	0.425
	(0.047)	(0.054)	(0.044)	(0.055)
Real GDP growth (%)	−0.097	−0.130	−0.229	
	(0.050)	(0.078)	(0.080)	
Current account (% of GDP)	−0.125	−0.083	0.028	
	(0.016)	(0.013)	(0.021)	
Year dummies	No	Yes	Yes	Yes
Country dummies	No	No	Yes	No
N	599	599	599	599
R^2	0.172	0.563	0.808	0.520

Note: Dependent variable: 10-year government bond yield (%). OLS on 599 country-years, 24 OECD countries, 2000–2024. Standard errors in parentheses: HC3 robust. Column (2) is the main model; column (1) drops the year dummies, column (3) adds country dummies, and column (4) drops real GDP growth and the current account balance, which debt may itself affect. The constant and the dummy coefficients are not shown.

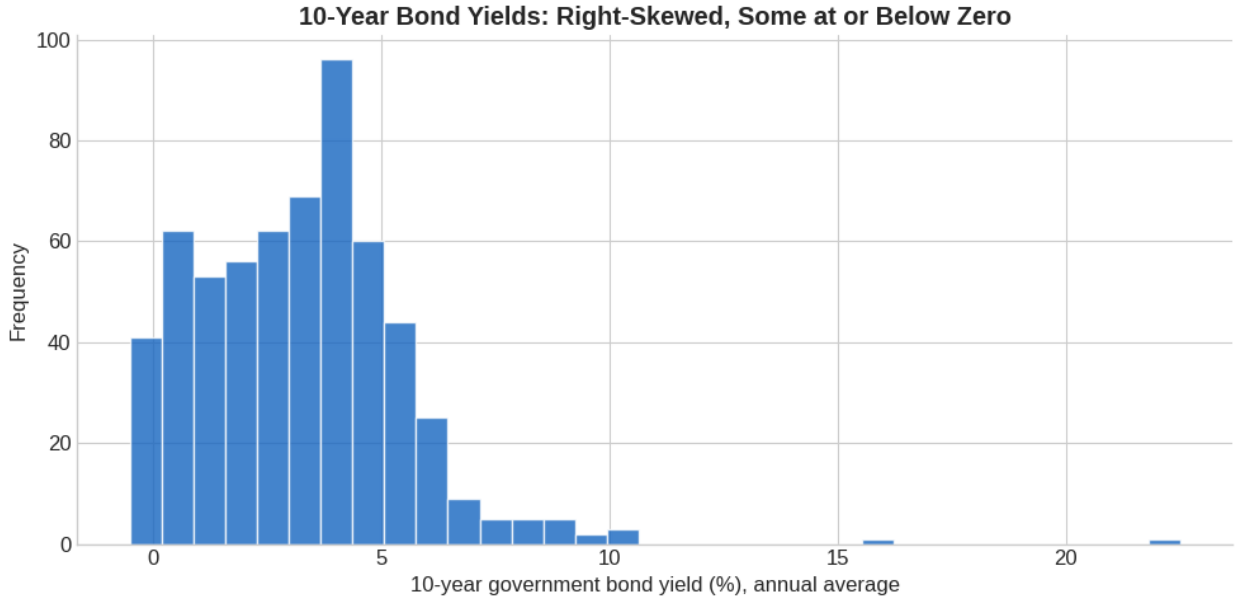


Figure A1. The yield is right-skewed, and 30 country-years are at or below zero. Histogram of the 599 annual yields, with Freedman–Diaconis bins; skewness 1.57. Source: OECD via FRED.

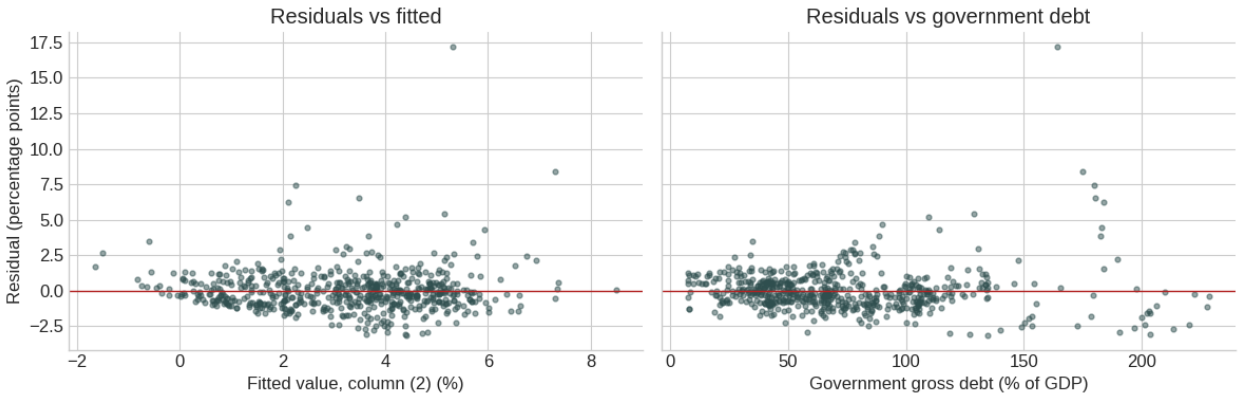


Figure A2. Residuals from column (2) against fitted values (left) and government debt (right). The spread widens with the fitted value: the residual standard deviation is 0.93 points in the lowest fifth of fitted values and 2.15 in the highest. The largest residual is Greece 2012 (standardised residual 11.5); Japan's residuals average −2.11 points.